Electric Vehicles
Charging Stations
Transformers
Charge Point Operator
Node Features
Power
Information
Pruned

(a) Common graph-structured PublicPST state

Node Features

Type-specific encoders

MLP
MLP
MLP
MLP

Initial node embeddings F_o

(b) Feature Extractor

Actor GNN backbone

Published reference: TD3 EV-GNN [9]

Controller variants introduced in this work

1. TD3 EV-GNN Baseline

EV action head

Shifted tanh

Direct EV actions

2. Full Hierarchical TD3 EV-GNN

Graph/CPO scale

Transformer allocation

Charger allocation

EV decision

EV actions

3. Hierarchy without Charger-Level Allocation

Graph/CPO scale

Transformer allocation

Charger allocation removed

Within-transformer EV allocation + EV decision

EV actions

4. Full Hierarchy with Hard Projection

Full Hierarchy (same as Variant 2)

Candidate EV actions

Hard transformer-feasibility projection

Projected EV actions

(c) Actor / action formation

Executed EV action

F_o + action → State-action GNN → Graph pooling → $Q(s_t, a_t)$

EV2Gym → r_t, s_{t+1}

(d) Environment execution and TD3 value evaluation